

Stat 565 Final Report

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Introduction

This report summarizes the completed replication and extension project for *Topological Data Analysis of Financial Time Series: Landscapes of Crashes* by Marian Gidea and Yuri Katz (<https://arxiv.org/pdf/1703.04385>). Following my proposal and intermediate report, the main objective is to reproduce the empirical pipeline for topological indicators based on persistence landscapes and to assess whether induced time series behave in an informative way prior to major U.S. equity stress episodes, benchmarked against standard volatility measures. In complementary terms, the question is whether multivariate geometry of recent returns carries incremental information relative to dispersion and dependence summaries alone.

The work has two components. First, I replicate the paper’s baseline U.S. implementation: Yahoo Finance inputs, rolling multivariate clouds, persistence landscapes ($w \in \{50, 100\}$), and anchored diagnostics around the dot-com and Lehman dates. Second, I apply identical code to an extended horizon through 2020, including the COVID disruption, so that conclusions can be compared across an episode outside the authors’ estimation window without changing the substantive procedure midway. Subsequent sections summarize the underlying construction, replication outputs, extension results, and limitations.

Background and motivation

At each date t , the construction fixes a backward-looking window of length w over d traded indices (here $d = 4$). Daily log-returns yield vectors in $\mathbb{R}^d$, so the trailing w observations form a point cloud $X_t \subset \mathbb{R}^d$ describing simultaneous index behavior rather than unrelated univariate traces. Means, variances, and correlations characterize much of X_t through first- and second-order structure; the motivating claim in Gidea and Katz is that regimes prior to crises may reorganize geometric features visible only when the full cloud is summarized at multiple spatial scales—for example redistribution of density or stable loop-like structure across resolutions—prior to unmistakable shocks in volatility on every coordinate. Persistent homology operationalizes qualitative “shape” in a repeatable fashion.

Roughly speaking, persistent homology analyzes connectivity of thickened neighborhoods (ϵ -neighborhood complexes) built from X_t as ϵ increases from zero. Focusing on homology in degree H_1 tracks topological loops: each surviving loop admits a lifetime between birth b and death d , yielding a persistence diagram $D_t^{(1)}$ of $\{(b, d)\}$ -pairs summarized by persistence landscapes $\lambda(\cdot)$; landscape norms $\|\lambda\|_1$ and $\|\lambda\|_2$ reduce this information to integrable scalars that update as t advances. The replicated implementation follows the authors’ H_1 / Vietoris–Rips emphasis and produces norms aligned alongside VIX rather than interpreting coordinates margin by margin alone.

Within that framework their intent is statistical early warning: systematically trending topological mass before anchored crash dates would corroborate multivariate deformation that classical aggregates underweight. Economic intuition is suggestive rather than strict: systemic stories of tightened linkages coincide qualitatively with geometric shifts in X_t . Because crisis mechanisms vary and histories are scarce, reproducibility exercises and disciplined out-of-sample extension remain necessary to separate robust behavior from illustrative windows; that is precisely what replication and COVID extrapolation furnish here.

Replication: Setup and Main Findings

Data and pipeline

The data source is Yahoo Finance (`yfinance`), corresponding to my replication notebook. Daily adjusted closes are retrieved for:

- S&P 500 (`^GSPC`)
- DJIA (`^DJI`)
- NASDAQ Composite (`^IXIC`)
- Russell 2000 (`^RUT`)

VIX (`^VIX`) enters as an external benchmark mirrored against landscape norms throughout, consistent with the paper’s juxtaposition against implied volatility. Computational steps coincide with their published sequencing:

1. Construct daily log-returns $r_{i,j} = \log(P_{i,j}/P_{i-1,j})$.
2. Compile rolling w -day clouds $X_t \subset \mathbb{R}^4$ for $w \in \{50, 100\}$.
3. Compute H_1 persistent homology with a Vietoris–Rips filtration using `ripser`.
4. Estimate persistence landscapes and record L^1, L^2 norms at each trailing window end-date.
5. Execute pre-anchor diagnostics (variance spectra, rolling variance, Kendall- τ trend tests aligned with interim course materials).

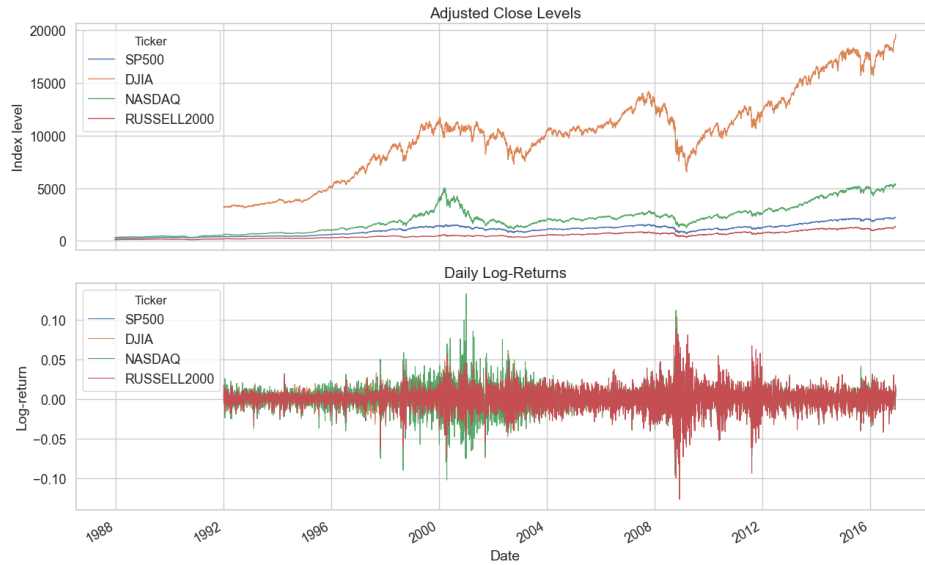


Figure 1: Replication data verification: adjusted close levels and aligned daily log-returns for the indices entering the topological pipeline.

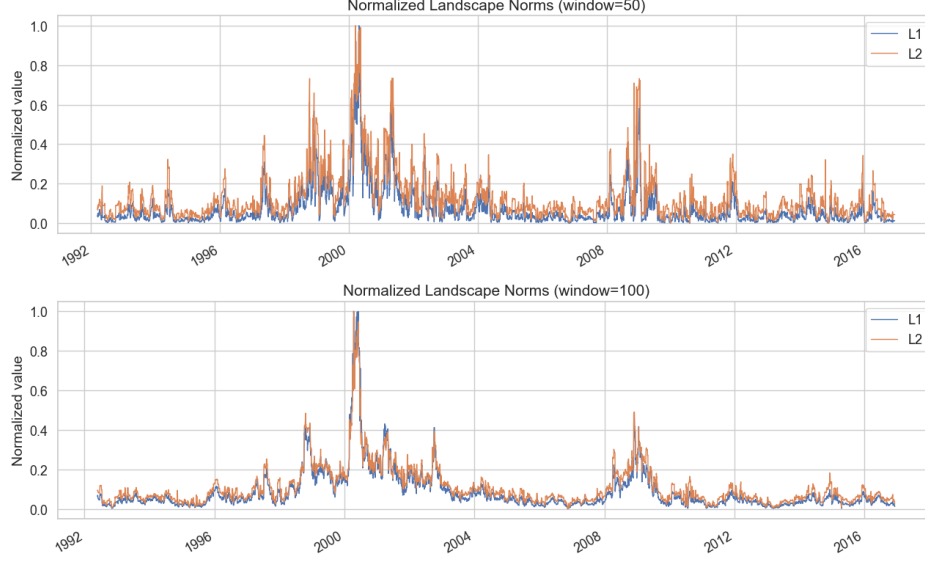


Figure 2: Persistence landscape norm trajectories (L^1 , L^2); elevated segments concentrate around stressed intervals.

Event-focused replication findings

Figure 3 repeats anchored comparisons for dot-com (left) and Lehman (right). Lehman aligns most cleanly with the paper’s emphasis on pre-event elevation: standardized L^1 evolves together with rising VIX in the buildup interval. Dot-com retains visible structure but behaves less uniformly across diagnostics, discouraging extrapolation from a uniform directional cue for every crisis solely from topology.

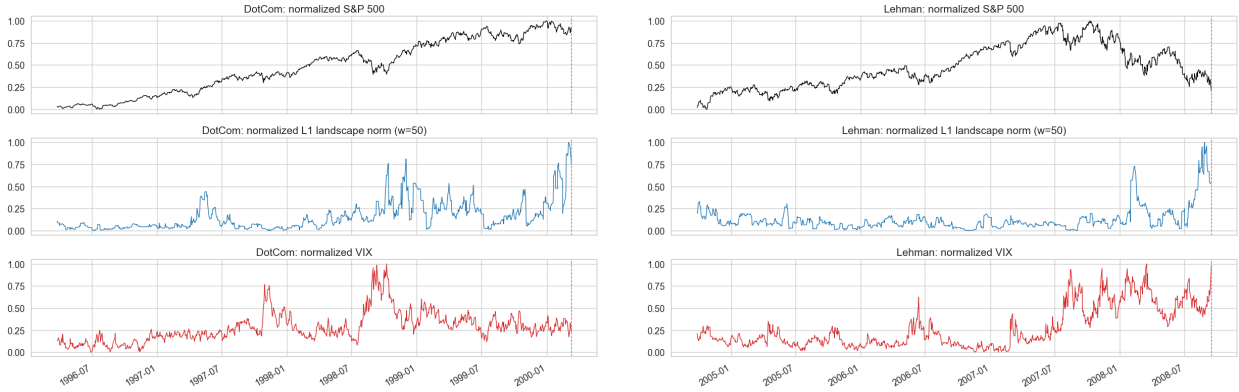


Figure 3: Event-centric panels: normalized S&P 500 trajectory, standardized L^1 ($w = 50$), and normalized VIX; dot-com versus Lehman anchors.

Table 1 summarizes Kendall- τ statistics (reported jointly with asymptotic p -values) applied to variance and low-frequency spectral components of landscape norms computed over 250 pre-anchor sessions. Lehman aligns directionally across both summaries with an escalating-risk portrayal; dot-com retains strong significance yet manifests opposing trend signs depending on spectral contrast, reaffirming event-specific heterogeneity emphasized graphically earlier.

Event	Variance trend (τ, p)	Low-frequency PSD trend (τ, p)
Dot-com	$-0.223, 1.44 \times 10^{-7}$	$-0.602, 1.42 \times 10^{-45}$
Lehman	$0.888, 4.36 \times 10^{-97}$	$0.691, 1.40 \times 10^{-59}$

Table 1: Kendall- τ trend tests on variance and low-frequency power of the landscape norm before each crash date.

Extension: Post-Paper Sample and COVID-Era Test

Extension design

The same codebase is rerun with only an extended Yahoo Finance pull through 2020 so that comparisons for the COVID episode do not hinge on discretionary changes to filtration or homology code introduced between samples.

Exported outputs mirror the replication bundle:

- persistence landscape norm series ($w = 50, w = 100$),
- aligned 4-index returns,
- VIX benchmark.

Post-paper behavior (2017 onward)

Figure 4 superimposes standardized landscape norms versus standardized VIX along the elongated sample path. Elevated turbulence intervals exhibit partial co-movement, yet divergence episodes remain visible, implying the topological signal conveys information beyond mechanical replication of implied volatility dynamics.

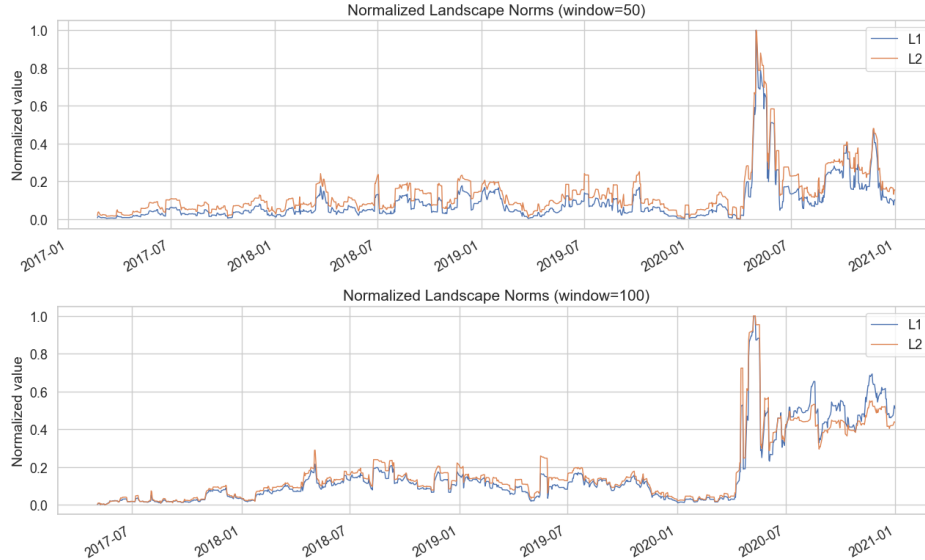


Figure 4: Post-replication horizon: juxtaposed standardized L^1 norms and standardized VIX benchmark.

COVID-era stress test

Because COVID lies outside Gidea and Katz’s published horizon, Figure 5 constitutes an out-of-sample check. Ahead of the February–March 2020 selloff, normalized L^1 remains near the bottom of the plotted

interval while VIX accelerates sharply, so this specification ($w = 50$; four-equity universe) affords limited justification for interpreting L^1 as an early-warning screen for COVID-specific stress.

Such a divergence is plausible when the shock transmits quickly through repricing tied to unfolding news rather than through the slower evolution of intra-window geometry summarized by homology on a fifty-day horizon. Robustness analyses (alternate window lengths, larger asset sets, diagnostics aimed at abrupt breaks) lie outside the present scope; nevertheless the pattern counsels restraint in generalizing unqualified early-warning claims beyond the historical episodes originally documented.

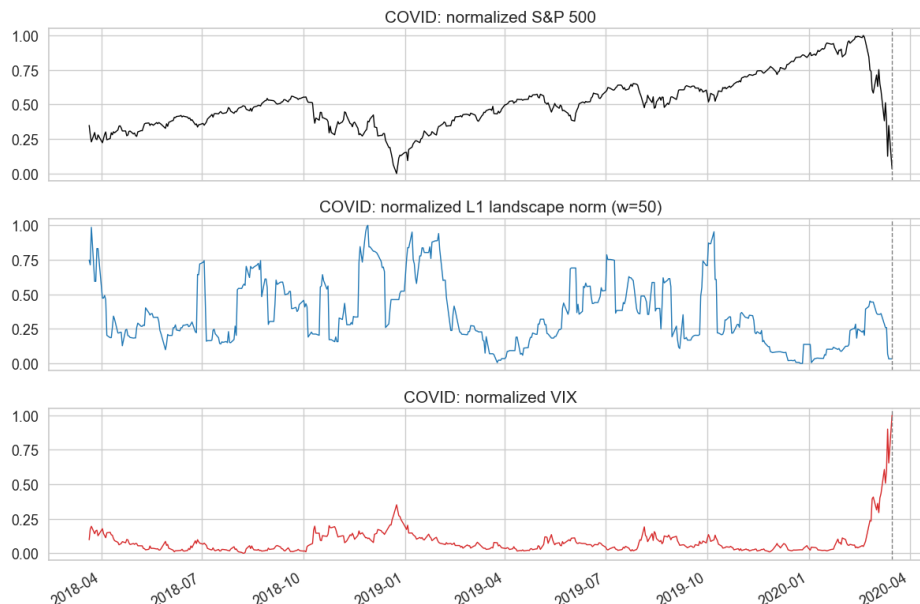


Figure 5: Focused COVID window: normalized aggregate index level alongside persistence landscape L^1 and VIX. Subdued L^1 preceding the downturn weakens preemptive signalling under this parametrization.

Conclusion

Replication demonstrates that persistence landscape norms can register familiar stress eras; congruence among figures and Kendall summaries is strongest for Lehman among the anchored cases reproduced here. Dot-com illustrates dispersion across diagnostics, implying that extrapolating one template to every episode would overstate uniformity.

Together with Figure 5, the extension indicates that analogous early-warning conclusions need not survive every newer shock absent retuning or complementary signals. Taken as a package, evidence supports interpreting topological trackers as exploratory complements to volatility benchmarks, calibrated case by case rather than asserted as universally leading predictors pending further empirical work.